Exploiting N-Terminal CSP Epitopes of Plasmodium Falciparum for their Immunogenicity Potential by Leveraging Site Directed Mutagenesis by PCR to Advance Malaria Vaccine Development

Author : Charité Igirimbabazi

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Abstract

This research proposal aims to elucidate the immunological role of the circumsporozoite protein's (CSP) N-terminal domain (ntCSP) in malaria, guiding a novel approach in vaccine research. Given the limited protection offered by the existing RTS,S/AS01 vaccine, with a 30%-40% efficacy (3, 4), this study seeks to uncover the ntCSP's potential as a critical target for immunological intervention (3). Through a scientific inquiry that includes the immunization of murine models, isolation of high-affinity antibodies, kinetic interaction analyses, and advanced molecular mapping techniques, the research is designed to provide a comprehensive understanding of the ntCSP's immunogenic properties. By employing methods such as ELISA for antibody identification, Surface Plasmon Resonance (SPR) for determining binding kinetics, cryo-Electron Microscopy (cryo-EM) for epitope visualization, and site-directed mutagenesis coupled with Next-Generation Sequencing (NGS) for variant analysis, this proposal aspires to dissect the ntCSP's role in eliciting an immune response. This focused exploration of the CSP N-terminal domain aims to pave the way for the development of more effective malaria vaccines by revealing novel immunological targets.

Introduction

Malaria, a disease caused by Plasmodium parasites, remains a global health challenge, largely due to its complex lifecycle which alternates between the Anopheles mosquito and the human host(1, 7, 15). The disease commences when sporozoites are transmitted into the human bloodstream via a mosquito bite, migrating to the liver to infect hepatocytes(1, 7). This pre-erythrocytic stage is crucial for the parasite's development and represents an optimal target for B-cell mediated immune strategies(1, 7). The circumsporozoite protein (CSP) is central to the parasite's ability to invade hepatocytes and is a prime candidate for vaccine development due to its surface exposure and role in host cell invasion(5, 9).

The RTS,S/AS01 vaccine, the only licensed malaria vaccine, utilizes the CSP's central repeat region and C-terminal but omits the N-terminal. It is composed of the CSP central repeat region (R), T-lymphocyte epitopes (T), and the Hepatitis B surface antigen (HBsAg, S) to elicit a broad immune response (8, 11, 14). While the vaccine has demonstrated approximately 30-40% efficacy (3, 4) in preventing clinical malaria, this partial protection underscores the necessity for new tools in malaria vaccine development. Notably, the N-terminal region of CSP has been underexplored in the formulation of existing vaccines(2), despite its crucial roles in the parasite's lifecycle and immune evasion strategies.

Figure1. N-terminal sequence of the circumsporozoite protein (CSP)

Recent studies have shown that the N-terminal region of CSP harbors unique characteristics, including the ability to bind C4b-binding protein, an apoptotic cell death promoter that is hijacked by pathogens for complement evasion(2, 5). This inhibition of the promoter reduces complement fixation and the subsequent inactivation of sporozoites in malaria infection. In addition, three distinct epitope regions within the CSP's N-terminal have been shown to elicit immune responses(2, 7). These regions are referred as ntCSP-A, ntCSP-B, and ntCSP-C, each recognized by specific monoclonal antibodies (mAbs). ntCSP-A, containing the Region I (RI) cysteine protease cleavage site(2), is crucial for the sporozoite's invasion into liver cells. RI's cleavage is hypothesized to be a requisite for hepatocyte entry, highlighting its importance in the parasite's lifecycle and as a target for immune intervention. In contrast, ntCSP-B, situated between amino acids 26-100, plays a role in complement evasion. The role of ntCSP-C is still not fully understood(2, 16, 17).

Given this immunological potential of the ntCSP, discovery of novel epitopes and characterization of these epitopes within the CSP N-terminal domain could redefine current paradigms in malaria vaccine

research(17), offering fresh targets for immunological intervention. This research proposal is dedicated to unraveling the molecular intricacies of the N-terminal domain of CSP and its repertoire of epitopes, both known (ntCSP-A, ntCSP-B, and ntCSP-C) and potentially undiscovered. By focusing on the N-terminal domain—a region hitherto less emphasized in vaccine development—this study posits that a deeper understanding of its associated epitopes could significantly advance our capability to design more effective malaria vaccines.

Aim 1: Immunogenic Profiling of ntCSP Epitopes in Murine Models

Since B-cell mediated immunity is good target to prevent sporozoites invasion and development in human hepatocytes, understanding we will carry immune response characterization with an emphasis on B-cell elicitation and specificity.

#1 Murine Model Immunization and Isolation of Highest binding antibodies

Synthetic ntCSP constructs will be acquired through a nucleotide manufacturer to initiate the targeted immunization regimen in murine models. The appropriate immunization protocol will be utilized to evoke a robust immune response. Post-immunization, we will employ an Enzyme-Linked Immunosorbent Assay (ELISA) to screen and quantify the antibody repertoire produced by the mice. The colorimetric ELISA functions on the principle of detecting and quantifying antigen-antibody interactions through a color change due to an enzyme linked to an antibody reaction with a chromogenic substrate. The intensity of the color produced will be employed as a quantitative medium to measure the amount of antibody bound to the ntCSP antigens on the plate. By this method, we will be able to identify and isolate the antibodies that demonstrate the highest binding capacities.

#2 SPR for kinetics of antibody-antigen interactions and use cryo-EM for epitopes mapping

We will apply Surface Plasmon Resonance (SPR) to analyze the kinetics of interactions between high-affinity antibodies and ntCSP antigens, identifying binding rates and affinities. The dissociation constants (Kd) will be recorded for each high-affinity antibody-antigen interactions. Complementarily, cryo-Electron Microscopy (cryo-EM) will be used for precise mapping of ntCSP epitopes, revealing the structural details of how antibodies recognize and bind to these antigens. This will reveal specific epitopes on the ntCSP that will be critical to guide our directed evolution studies.

If ELISA results indicate low antibody affinity or diversity, we'll consider revising the immunization strategy by adjusting antigen dosage or exploring different adjuvants.

Aim 2: Refinement of ntCSP Epitope Variants Through Site-Directed Mutagenesis

Building upon the insights gathered from immunogenic profiling in murine models, Aim 2 seeks to harness the precision of site-directed mutagenesis to construct a library of ntCSP epitope variants. This is critical in investigating key residues involved in the immunogenicity role of the ntCSP.

#1 Library of variants generation through site directed mutations specifically in the identified ntCSP epitopes

A suite of primers, each tailored to introduce specific mutations within the immunogenic regions of ntCSP epitopes will be designed then acquired. These primers that will introduce targeted mutations into the DNA sequence of ntCSP epitopes through extension PCR. This will generate a diverse collection if ntCSP variants. The application of this technique will ensure a focused exploration of structural variations and their immunological consequences.

#2 Transformation and sequencing to verify the integration of mutations

Following the creation of ntCSP epitope variants, we will proceed with their transformation into an appropriate bacterial expression system. This step facilitates the propagation of the mutant constructs and prepares them for subsequent validation. After the transformation process, we'll isolate DNA from the transformed hosts and apply Next-Generation Sequencing (NGS) to verify the integration of our targeted mutations within the ntCSP epitopes. NGS offers comprehensive coverage and high-throughput capabilities, making it ideal for confirming the presence and precision of site-directed mutagenesis across a broad array of ntCSP variants.

If NGS reveals inaccuracies in the ntCSP epitope mutations, we will refine our primer designs and optimize transformation protocols to correct these discrepancies.

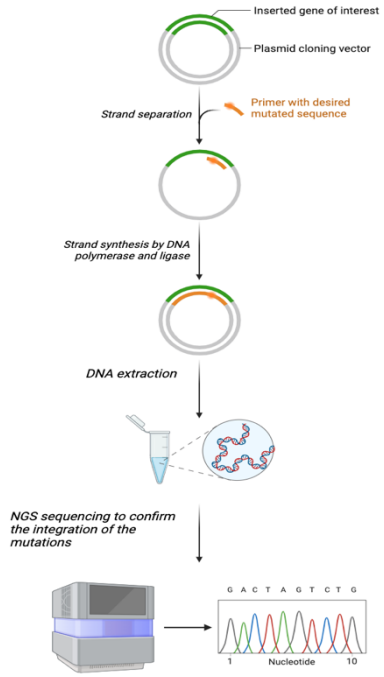

Figure2. Experimental set-up for transformation and sequencing for ntCSP mutants

Aim 3: Molecular Characterization of ntCSP-Antibody Binding Dynamics

Aim 3 will undertake a comparative biophysical examination of antibody interactions with both the wild-type and mutant ntCSP epitopes, quantifying the effects of epitope variations on antibody affinity. We will delve into the differential equilibrium dissociation constants (Kd) for ntCSP-antibody complexes, comparing the binding efficiency of antibodies with wild-type ntCSP to that of the mutant variants.

#1 SPR differential binding analysis for binding affinity measurement between wild type ntCSP and mutants ntCSP

Utilizing Surface Plasmon Resonance (SPR) for its sensitivity and specificity, we will conduct a comparative binding analysis to measure and contrast the Kd values of antibody complexes with wild-type and mutant ntCSP epitopes. By quantifying the Kd values, we will gain insight into how amino acid changes within the epitopes influence antibody recognition and binding.

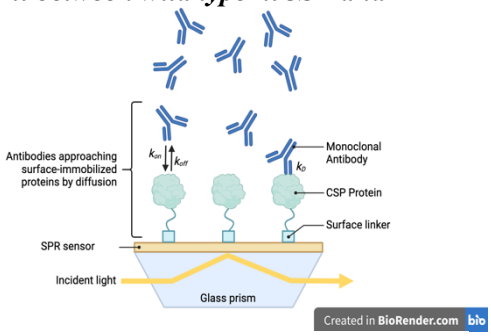

Figure3. SPR set-up to measure the binding affinity of ntCSP-antibody complex

#2 Cryo-EM conformational analysis

Concurrently, Cryo-Electron Microscopy (cryo-EM) will be employed to perform a structural analysis of the ntCSP-antibody complexes, allowing us to visualize the precise conformational details of epitope-antibody interactions. The high-resolution images obtained through cryo-EM are instrumental for correlating the binding affinity data gathered via SPR with structural modifications in the epitope. This correlation is invaluable for pinpointing the structural basis of enhanced or diminished antibody binding and guides the optimization of epitope design for vaccine development.

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